SPENVIS 4.6.14.3582 10-May-2026 02:25:31

Solar cell damage equivalent fluences

Project: GEO_10Y

geo_10yers_duration

Contents

Cell: Azur 3G30 (SR-NIEL 21eV)
Electron/proton damage ratios: P_{max} = 1134.8 V_{oc} = 2864.6 I_{sc} = 613.4
Coverglass
material: $SiO2$ density: 2.32 (g/cm^3)

Summary of 1 MeV equivalent electron fluences (cm^{-2})

Coverglass thickness			Total			Trapped electrons	Trapped protons			Solar protons		
g cm^{-2}	mils	micron	P_{max}	V_{oc}	I_{sc}	P_{max}, V_{oc}, I_{sc}	P_{max}	V_{oc}	I_{sc}	P_{max}	V_{oc}	I_{sc}
0.0000	0.0	0.00	1.292E+19	3.260E+19	6.982E+18	1.627E+14	1.285E+19	3.245E+19	6.948E+18	6.248E+16	1.577E+17	3.378E+16
0.0059	1.0	25.40	2.478E+15	6.039E+15	1.405E+15	1.419E+14	1.226E+15	3.096E+15	6.630E+14	1.110E+15	2.801E+15	5.998E+14
0.0177	3.0	76.20	5.715E+14	1.261E+15	3.637E+14	1.193E+14	3.731E+13	9.417E+13	2.017E+13	4.149E+14	1.047E+15	2.243E+14
0.0354	6.0	152.40	3.194E+14	6.571E+14	2.176E+14	9.784E+13	1.951E+11	4.926E+11	1.055E+11	2.213E+14	5.587E+14	1.196E+14
0.0707	12.0	304.80	1.821E+14	3.517E+14	1.310E+14	7.085E+13	6.707E+07	1.693E+08	3.625E+07	1.113E+14	2.808E+14	6.014E+13
0.1179	20.0	508.00	1.184E+14	2.228E+14	8.696E+13	4.994E+13	1.000E-12	1.000E-12	1.000E-12	6.849E+13	1.729E+14	3.702E+13
0.1768	30.0	762.00	8.064E+13	1.516E+14	5.925E+13	3.408E+13	1.000E-12	1.000E-12	1.000E-12	4.656E+13	1.175E+14	2.517E+13
0.3536	60.0	1524.00	3.376E+13	6.568E+13	2.414E+13	1.282E+13	1.000E-12	1.000E-12	1.000E-12	2.094E+13	5.286E+13	1.132E+13
User defined coverglass thickness												
0.0232	3.9	100.00	4.549E+14	9.764E+14	2.973E+14	1.104E+14	4.755E+12	1.200E+13	2.570E+12	3.243E+14	8.186E+14	1.753E+14

Trapped protons for mission segment 1 for P_{max} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	7.209E+14	4.643E+14	1.85E+01	8.591E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.20	0.30	2.566E+14	1.530E+14	1.23E+01	1.881E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.30	0.40	1.036E+14	6.938E+13	9.27E+00	6.431E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.40	0.60	3.421E+13	2.454E+13	6.91E+00	1.695E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.60	0.80	9.673E+12	5.631E+12	5.02E+00	2.829E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.80	1.00	4.042E+12	2.126E+12	4.01E+00	8.527E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.00	1.20	1.916E+12	7.777E+11	3.34E+00	2.597E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.20	1.30	1.139E+12	2.326E+11	2.95E+00	6.867E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.30	1.40	9.061E+11	1.728E+11	2.75E+00	4.754E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000

1.40	1.60	7.333E+11	3.369E+11	2.49E+00	8.383E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.60	1.80	3.964E+11	2.116E+11	2.25E+00	4.768E+11	2.88E+00	6.104E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.80	2.00	1.848E+11	9.143E+10	2.04E+00	1.862E+11	2.87E+00	2.620E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.00	2.20	9.340E+10	2.569E+10	1.85E+00	4.758E+10	2.67E+00	6.867E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.20	2.40	6.771E+10	1.723E+10	1.71E+00	2.940E+10	2.46E+00	4.242E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.40	2.60	5.048E+10	1.195E+10	1.59E+00	1.895E+10	2.26E+00	2.703E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.60	2.80	3.853E+10	8.526E+09	1.48E+00	1.263E+10	2.08E+00	1.770E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.80	3.00	3.000E+10	6.232E+09	1.39E+00	8.636E+09	1.92E+00	1.197E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.00	3.20	2.377E+10	1.458E+10	1.31E+00	1.909E+10	1.79E+00	2.606E+10	1.33E+00	1.939E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.20	3.40	9.184E+09	5.425E+09	1.24E+00	6.718E+09	1.66E+00	8.982E+09	1.47E+00	7.985E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.40	3.60	3.760E+09	2.140E+09	1.17E+00	2.510E+09	1.54E+00	3.305E+09	1.50E+00	3.200E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	1.620E+09	8.894E+08	1.11E+00	9.896E+08	1.45E+00	1.289E+09	1.48E+00	1.313E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	7.302E+08	3.873E+08	1.06E+00	4.094E+08	1.36E+00	5.280E+08	1.44E+00	5.566E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.430E+08	8.883E+07	1.01E+00	8.973E+07	1.29E+00	1.142E+08	1.38E+00	1.227E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	2.542E+08	6.318E+07	9.70E-01	6.129E+07	1.22E+00	7.682E+07	1.33E+00	8.376E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	1.910E+08	4.564E+07	9.32E-01	4.256E+07	1.15E+00	5.252E+07	1.28E+00	5.822E+07	7.34E-01	3.350E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	1.453E+08	3.344E+07	8.98E-01	3.003E+07	1.10E+00	3.664E+07	1.22E+00	4.082E+07	8.95E-01	2.994E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	1.119E+08	7.578E+07	8.46E-01	6.410E+07	1.02E+00	7.706E+07	1.14E+00	8.611E+07	9.70E-01	7.352E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	3.612E+07	2.927E+07	8.02E-01	2.346E+07	9.50E-01	2.781E+07	1.06E+00	3.112E+07	9.85E-01	2.883E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	6.848E+06	5.392E+06	7.54E-01	4.064E+06	8.80E-01	4.744E+06	9.81E-01	5.289E+06	9.60E-01	5.178E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	1.456E+06	6.413E+05	7.11E-01	4.560E+05	8.17E-01	5.240E+05	9.10E-01	5.835E+05	9.17E-01	5.883E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	8.151E+05	3.426E+05	6.77E-01	2.319E+05	7.66E-01	2.625E+05	8.50E-01	2.912E+05	8.73E-01	2.990E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	4.725E+05	4.725E+05	1.48E-01	6.989E+04	1.66E-01	7.837E+04	1.83E-01	8.661E+04	1.90E-01	8.975E+04	1.25E-01	5.910E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
12.00	13.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
13.00	14.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
14.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
15.00	16.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
16.00	18.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
18.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
20.00	22.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
22.00	24.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
24.00	26.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
26.00	28.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
28.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
30.00	34.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
34.00	38.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
38.00	42.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
42.00	46.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
46.00	50.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
50.00	55.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
55.00	60.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
60.00	65.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
65.00	70.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
70.00	80.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
80.00	90.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
90.00	100.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
100.00	130.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
130.00	160.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
160.00	200.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
Equiv. 10 MeV Proton Fluence (cm^{-2})					1.133E+16		1.081E+12		3.288E+10		1.719E+08		5.910E+04		0.000E+00		0.000E+00		0.00
P_{max} Equiv. 1 MeV Electron Fluence (cm^{-2})					1.285E+19		1.226E+15		3.731E+13		1.951E+11		6.707E+07		0.000E+00		0.000E+00		0.00

Trapped protons for mission segment 1 for V_{oc} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	7.209E+14	4.643E+14	1.85E+01	8.591E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.20	0.30	2.566E+14	1.530E+14	1.23E+01	1.881E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.30	0.40	1.036E+14	6.938E+13	9.27E+00	6.431E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.40	0.60	3.421E+13	2.454E+13	6.91E+00	1.695E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.60	0.80	9.673E+12	5.631E+12	5.02E+00	2.829E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.80	1.00	4.042E+12	2.126E+12	4.01E+00	8.527E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00

1.00	1.20	1.916E+12	7.777E+11	3.34E+00	2.597E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.20	1.30	1.139E+12	2.326E+11	2.95E+00	6.867E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.30	1.40	9.061E+11	1.728E+11	2.75E+00	4.754E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.40	1.60	7.333E+11	3.369E+11	2.49E+00	8.383E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.60	1.80	3.964E+11	2.116E+11	2.25E+00	4.768E+11	2.88E+00	6.104E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.80	2.00	1.848E+11	9.143E+10	2.04E+00	1.862E+11	2.87E+00	2.620E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.00	2.20	9.340E+10	2.569E+10	1.85E+00	4.758E+10	2.67E+00	6.867E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.20	2.40	6.771E+10	1.723E+10	1.71E+00	2.940E+10	2.46E+00	4.242E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.40	2.60	5.048E+10	1.195E+10	1.59E+00	1.895E+10	2.26E+00	2.703E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.60	2.80	3.853E+10	8.526E+09	1.48E+00	1.263E+10	2.08E+00	1.770E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.80	3.00	3.000E+10	6.232E+09	1.39E+00	8.636E+09	1.92E+00	1.197E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.00	3.20	2.377E+10	1.458E+10	1.31E+00	1.909E+10	1.79E+00	2.606E+10	1.33E+00	1.939E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.20	3.40	9.184E+09	5.425E+09	1.24E+00	6.718E+09	1.66E+00	8.982E+09	1.47E+00	7.985E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.40	3.60	3.760E+09	2.140E+09	1.17E+00	2.510E+09	1.54E+00	3.305E+09	1.50E+00	3.200E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	1.620E+09	8.894E+08	1.11E+00	9.896E+08	1.45E+00	1.289E+09	1.48E+00	1.313E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	7.302E+08	3.873E+08	1.06E+00	4.094E+08	1.36E+00	5.280E+08	1.44E+00	5.566E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.430E+08	8.883E+07	1.01E+00	8.973E+07	1.29E+00	1.142E+08	1.38E+00	1.227E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	2.542E+08	6.318E+07	9.70E-01	6.129E+07	1.22E+00	7.682E+07	1.33E+00	8.376E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	1.910E+08	4.564E+07	9.32E-01	4.256E+07	1.15E+00	5.252E+07	1.28E+00	5.822E+07	7.34E-01	3.350E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	1.453E+08	3.344E+07	8.98E-01	3.003E+07	1.10E+00	3.664E+07	1.22E+00	4.082E+07	8.95E-01	2.994E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	1.119E+08	7.578E+07	8.46E-01	6.410E+07	1.02E+00	7.706E+07	1.14E+00	8.611E+07	9.70E-01	7.352E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	3.612E+07	2.927E+07	8.02E-01	2.346E+07	9.50E-01	2.781E+07	1.06E+00	3.112E+07	9.85E-01	2.883E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	6.848E+06	5.392E+06	7.54E-01	4.064E+06	8.80E-01	4.744E+06	9.81E-01	5.289E+06	9.60E-01	5.178E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	1.456E+06	6.413E+05	7.11E-01	4.560E+05	8.17E-01	5.240E+05	9.10E-01	5.835E+05	9.17E-01	5.883E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	8.151E+05	3.426E+05	6.77E-01	2.319E+05	7.66E-01	2.625E+05	8.50E-01	2.912E+05	8.73E-01	2.990E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	4.725E+05	4.725E+05	1.48E-01	6.989E+04	1.66E-01	7.837E+04	1.83E-01	8.661E+04	1.90E-01	8.975E+04	1.25E-01	5.910E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
12.00	13.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
13.00	14.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
14.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
15.00	16.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
16.00	18.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
18.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
20.00	22.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
22.00	24.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
24.00	26.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
26.00	28.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
28.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
30.00	34.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
34.00	38.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
38.00	42.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
42.00	46.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
46.00	50.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
50.00	55.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
55.00	60.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
60.00	65.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
65.00	70.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
70.00	80.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
80.00	90.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
90.00	100.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
100.00	130.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
130.00	160.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
160.00	200.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
Equiv. 10 MeV Proton Fluence (cm^{-2})					1.133E+16		1.081E+12		3.288E+10		1.719E+08		5.910E+04		0.000E+00		0.000E+00		0.00
V_{OC} Equiv. 1 MeV Electron Fluence (cm^{-2})					3.245E+19		3.096E+15		9.417E+13		4.926E+11		1.693E+08		0.000E+00		0.000E+00		0.00

Trapped protons for mission segment 1 for I_{SC} (integration restricted to the input energy range). D(E,..) values are coarse averages introdu into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	7.209E+14	4.643E+14	1.85E+01	8.591E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.20	0.30	2.566E+14	1.530E+14	1.23E+01	1.881E+15	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.30	0.40	1.036E+14	6.938E+13	9.27E+00	6.431E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.40	0.60	3.421E+13	2.454E+13	6.91E+00	1.695E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00

0.60	0.80	9.673E+12	5.631E+12	5.02E+00	2.829E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
0.80	1.00	4.042E+12	2.126E+12	4.01E+00	8.527E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.00	1.20	1.916E+12	7.777E+11	3.34E+00	2.597E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.20	1.30	1.139E+12	2.326E+11	2.95E+00	6.867E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.30	1.40	9.061E+11	1.728E+11	2.75E+00	4.754E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.40	1.60	7.333E+11	3.369E+11	2.49E+00	8.383E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.60	1.80	3.964E+11	2.116E+11	2.25E+00	4.768E+11	2.88E+00	6.104E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
1.80	2.00	1.848E+11	9.143E+10	2.04E+00	1.862E+11	2.87E+00	2.620E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
2.00	2.20	9.340E+10	2.569E+10	1.85E+00	4.758E+10	2.67E+00	6.867E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
2.20	2.40	6.771E+10	1.723E+10	1.71E+00	2.940E+10	2.46E+00	4.242E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
2.40	2.60	5.048E+10	1.195E+10	1.59E+00	1.895E+10	2.26E+00	2.703E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
2.60	2.80	3.853E+10	8.526E+09	1.48E+00	1.263E+10	2.08E+00	1.770E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
2.80	3.00	3.000E+10	6.232E+09	1.39E+00	8.636E+09	1.92E+00	1.197E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
3.00	3.20	2.377E+10	1.458E+10	1.31E+00	1.909E+10	1.79E+00	2.606E+10	1.33E+00	1.939E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
3.20	3.40	9.184E+09	5.425E+09	1.24E+00	6.718E+09	1.66E+00	8.982E+09	1.47E+00	7.985E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
3.40	3.60	3.760E+09	2.140E+09	1.17E+00	2.510E+09	1.54E+00	3.305E+09	1.50E+00	3.200E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
3.60	3.80	1.620E+09	8.894E+08	1.11E+00	9.896E+08	1.45E+00	1.289E+09	1.48E+00	1.313E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
3.80	4.00	7.302E+08	3.873E+08	1.06E+00	4.094E+08	1.36E+00	5.280E+08	1.44E+00	5.566E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
4.00	4.20	3.430E+08	8.883E+07	1.01E+00	8.973E+07	1.29E+00	1.142E+08	1.38E+00	1.227E+08	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
4.20	4.40	2.542E+08	6.318E+07	9.70E-01	6.129E+07	1.22E+00	7.682E+07	1.33E+00	8.376E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
4.40	4.60	1.910E+08	4.564E+07	9.32E-01	4.256E+07	1.15E+00	5.252E+07	1.28E+00	5.822E+07	7.34E-01	3.350E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
4.60	4.80	1.453E+08	3.344E+07	8.98E-01	3.003E+07	1.10E+00	3.664E+07	1.22E+00	4.082E+07	8.95E-01	2.994E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
4.80	5.20	1.119E+08	7.578E+07	8.46E-01	6.410E+07	1.02E+00	7.706E+07	1.14E+00	8.611E+07	9.70E-01	7.352E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
5.20	5.60	3.612E+07	2.927E+07	8.02E-01	2.346E+07	9.50E-01	2.781E+07	1.06E+00	3.112E+07	9.85E-01	2.883E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
5.60	6.00	6.848E+06	5.392E+06	7.54E-01	4.064E+06	8.80E-01	4.744E+06	9.81E-01	5.289E+06	9.60E-01	5.178E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
6.00	6.40	1.456E+06	6.413E+05	7.11E-01	4.560E+05	8.17E-01	5.240E+05	9.10E-01	5.835E+05	9.17E-01	5.883E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
6.40	6.80	8.151E+05	3.426E+05	6.77E-01	2.319E+05	7.66E-01	2.625E+05	8.50E-01	2.912E+05	8.73E-01	2.990E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
6.80	7.20	4.725E+05	4.725E+05	1.48E-01	6.989E+04	1.66E-01	7.837E+04	1.83E-01	8.661E+04	1.90E-01	8.975E+04	1.25E-01	5.910E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
7.20	7.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
7.60	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
10.00	11.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
11.00	12.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
12.00	13.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
13.00	14.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
14.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
15.00	16.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
16.00	18.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
18.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
20.00	22.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
22.00	24.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
24.00	26.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
26.00	28.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
28.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
30.00	34.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
34.00	38.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
38.00	42.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
42.00	46.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
46.00	50.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
50.00	55.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
55.00	60.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
60.00	65.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
65.00	70.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
70.00	80.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
80.00	90.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
90.00	100.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
100.00	130.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
130.00	160.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
160.00	200.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00(
Equiv. 10 MeV Proton Fluence (cm^{-2})					1.133E+16		1.081E+12		3.288E+10		1.719E+08		5.910E+04		0.000E+00		0.000E+00		0.00(
I_{SC} Equiv. 1 MeV Electron Fluence (cm^{-2})					6.948E+18		6.630E+14		2.017E+13		1.055E+11		3.625E+07		0.000E+00		0.000E+00		0.00(

Trapped electrons for mission segment 1 (integration restricted to the input energy range). D(E,..) values are coarse averages introduce into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Fluer (cm^{-2}
0.15	0.16	3.880E+15	2.759E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.16	0.17	3.604E+15	2.563E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-

0.17	0.18	3.348E+15	2.381E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.18	0.19	3.109E+15	2.211E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.19	0.20	2.888E+15	2.054E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.20	0.22	2.683E+15	2.879E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.22	0.24	2.395E+15	2.570E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.24	0.26	2.138E+15	2.294E+14	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.26	0.28	1.909E+15	2.048E+14	6.20E-05	1.269E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.28	0.30	1.704E+15	1.828E+14	2.01E-04	3.671E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.30	0.32	1.521E+15	1.571E+14	3.60E-04	5.650E+10	1.56E-04	2.449E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.32	0.36	1.364E+15	2.672E+14	8.11E-04	2.167E+11	3.78E-04	1.010E+11	8.18E-05	2.184E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.36	0.40	1.097E+15	2.148E+14	2.13E-03	4.581E+11	1.08E-03	2.317E+11	4.11E-04	8.831E+10	6.91E-05	1.485E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.40	0.45	8.820E+14	1.923E+14	5.68E-03	1.092E+12	3.09E-03	5.935E+11	1.31E-03	2.525E+11	4.10E-04	7.880E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.45	0.50	6.897E+14	1.504E+14	1.49E-02	2.245E+12	8.65E-03	1.301E+12	4.06E-03	6.102E+11	1.47E-03	2.205E+11	1.83E-04	2.759E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.50	0.60	5.393E+14	1.726E+14	5.45E-02	9.407E+12	3.40E-02	5.875E+12	1.78E-02	3.072E+12	7.58E-03	1.309E+12	1.43E-03	2.467E+11	9.54E-05	1.647E+10	0.00E+00	0.000E+00	0.00E+00	0.000E-
0.60	0.70	3.667E+14	1.178E+14	1.56E-01	1.837E+13	1.13E-01	1.331E+13	7.21E-02	8.493E+12	3.96E-02	4.666E+12	1.11E-02	1.309E+12	1.64E-03	1.938E+11	7.39E-05	8.708E+09	0.00E+00	0.000E-
0.70	0.80	2.489E+14	6.594E+13	2.55E-01	1.683E+13	2.13E-01	1.401E+13	1.62E-01	1.068E+13	1.11E-01	7.320E+12	4.70E-02	3.102E+12	1.20E-02	7.944E+11	1.33E-03	8.773E+10	0.00E+00	0.000E-
0.80	0.90	1.829E+14	4.318E+13	3.54E-01	1.530E+13	3.10E-01	1.339E+13	2.54E-01	1.097E+13	1.94E-01	8.388E+12	1.11E-01	4.814E+12	4.63E-02	1.999E+12	9.98E-03	4.311E+11	0.00E+00	0.000E-
0.90	1.00	1.398E+14	3.299E+13	4.52E-01	1.491E+13	4.07E-01	1.341E+13	3.47E-01	1.144E+13	2.81E-01	9.259E+12	1.85E-01	6.100E+12	1.01E-01	3.346E+12	3.81E-02	1.257E+12	7.18E-05	2.369E-
1.00	1.20	1.068E+14	3.882E+13	5.60E-01	2.172E+13	5.19E-01	2.014E+13	4.62E-01	1.793E+13	3.95E-01	1.534E+13	2.92E-01	1.134E+13	1.92E-01	7.466E+12	1.03E-01	4.014E+12	3.47E-03	1.348E-
1.20	1.40	6.795E+13	2.539E+13	6.92E-01	1.757E+13	6.55E-01	1.664E+13	6.03E-01	1.530E+13	5.39E-01	1.369E+13	4.36E-01	1.108E+13	3.28E-01	8.340E+12	2.22E-01	5.633E+12	3.87E-02	9.821E-
1.40	1.60	4.256E+13	1.600E+13	8.29E-01	1.327E+13	7.91E-01	1.265E+13	7.36E-01	1.178E+13	6.70E-01	1.072E+13	5.64E-01	9.026E+12	4.52E-01	7.239E+12	3.42E-01	5.465E+12	1.15E-01	1.842E-
1.60	1.80	2.656E+13	1.000E+13	9.72E-01	9.726E+12	9.32E-01	9.325E+12	8.74E-01	8.746E+12	8.04E-01	8.046E+12	6.93E-01	6.927E+12	5.74E-01	5.742E+12	4.56E-01	4.556E+12	2.07E-01	2.074E-
1.80	2.00	1.655E+13	6.318E+12	1.12E+00	7.073E+12	1.08E+00	6.820E+12	1.01E+00	6.412E+12	9.44E-01	5.962E+12	8.25E-01	5.210E+12	6.98E-01	4.412E+12	5.71E-01	3.607E+12	3.02E-01	1.905E-
2.00	2.25	1.024E+13	4.430E+12	1.26E+00	5.562E+12	1.22E+00	5.405E+12	1.15E+00	5.115E+12	1.09E+00	4.824E+12	9.68E-01	4.286E+12	8.38E-01	3.714E+12	7.02E-01	3.112E+12	4.07E-01	1.805E-
2.25	2.50	5.806E+12	2.564E+12	1.38E+00	3.527E+12	1.35E+00	3.450E+12	1.29E+00	3.307E+12	1.23E+00	3.142E+12	1.11E+00	2.845E+12	9.86E-01	2.529E+12	8.48E-01	2.175E+12	5.31E-01	1.362E-
2.50	2.75	3.242E+12	1.321E+12	1.49E+00	1.970E+12	1.46E+00	1.930E+12	1.41E+00	1.864E+12	1.35E+00	1.777E+12	1.24E+00	1.632E+12	1.12E+00	1.473E+12	9.83E-01	1.298E+12	6.61E-01	8.726E-
2.75	3.00	1.921E+12	7.618E+11	1.61E+00	1.224E+12	1.58E+00	1.201E+12	1.53E+00	1.163E+12	1.47E+00	1.117E+12	1.36E+00	1.033E+12	1.24E+00	9.414E+11	1.11E+00	8.423E+11	7.86E-01	5.986E-
3.00	3.25	1.160E+12	4.175E+11	1.71E+00	7.131E+11	1.68E+00	7.025E+11	1.63E+00	6.816E+11	1.58E+00	6.585E+11	1.47E+00	6.145E+11	1.35E+00	5.644E+11	1.22E+00	5.100E+11	9.01E-01	3.761E-
3.25	3.50	7.421E+11	2.687E+11	1.79E+00	4.816E+11	1.77E+00	4.749E+11	1.72E+00	4.627E+11	1.67E+00	4.493E+11	1.57E+00	4.224E+11	1.46E+00	3.914E+11	1.33E+00	3.565E+11	1.01E+00	2.703E-
3.50	3.75	4.735E+11	1.535E+11	1.88E+00	2.883E+11	1.85E+00	2.837E+11	1.81E+00	2.776E+11	1.76E+00	2.699E+11	1.66E+00	2.553E+11	1.55E+00	2.384E+11	1.43E+00	2.191E+11	1.11E+00	1.707E-
3.75	4.00	3.200E+11	9.667E+10	1.96E+00	1.893E+11	1.93E+00	1.869E+11	1.89E+00	1.830E+11	1.84E+00	1.777E+11	1.75E+00	1.690E+11	1.64E+00	1.584E+11	1.52E+00	1.468E+11	1.21E+00	1.172E-
4.00	4.50	2.233E+11	1.015E+11	2.06E+00	2.090E+11	2.04E+00	2.070E+11	2.00E+00	2.034E+11	1.95E+00	1.977E+11	1.86E+00	1.886E+11	1.75E+00	1.779E+11	1.64E+00	1.661E+11	1.34E+00	1.356E-
4.50	5.00	1.218E+11	4.554E+10	2.19E+00	9.971E+10	2.17E+00	9.880E+10	2.14E+00	9.743E+10	2.09E+00	9.515E+10	2.00E+00	9.126E+10	1.90E+00	8.670E+10	1.79E+00	8.169E+10	1.50E+00	6.842E-
5.00	5.50	7.624E+10	2.340E+10	2.31E+00	5.412E+10	2.29E+00	5.365E+10	2.26E+00	5.295E+10	2.22E+00	5.189E+10	2.14E+00	5.002E+10	2.04E+00	4.779E+10	1.93E+00	4.521E+10	1.66E+00	3.877E-
5.50	6.00	5.283E+10	1.450E+10	2.42E+00	3.513E+10	2.40E+00	3.484E+10	2.37E+00	3.440E+10	2.33E+00	3.375E+10	2.25E+00	3.266E+10	2.16E+00	3.136E+10	2.06E+00	2.983E+10	1.79E+00	2.598E-
6.00	7.00	3.834E+10	3.834E+10	1.17E+00	4.468E+10	1.16E+00	4.433E+10	1.14E+00	4.381E+10	1.12E+00	4.302E+10	1.09E+00	4.188E+10	1.05E+00	4.031E+10	1.01E+00	3.864E+10	8.92E-01	3.418E-
7.00	8.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
8.00	9.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
9.00	10.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
10.00	15.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
15.00	20.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
20.00	25.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
25.00	30.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
30.00	40.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E-
Equiv. 1 MeV Electron Fluence (cm^{-2})					1.627E+14		1.419E+14		1.193E+14		9.784E+13		7.085E+13		4.994E+13		3.408E+13		1.282E-

Solar protons for P_{max} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table tc simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	4.895E+12	1.926E+12	1.85E+01	3.555E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.20	0.30	2.969E+12	7.532E+11	1.22E+01	9.156E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.30	0.40	2.216E+12	4.155E+11	9.15E+00	3.801E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.40	0.60	1.801E+12	4.564E+11	6.81E+00	3.108E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.60	0.80	1.344E+12	2.523E+11	5.00E+00	1.262E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
0.80	1.00	1.092E+12	1.621E+11	3.98E+00	6.459E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.00	1.20	9.299E+11	1.140E+11	3.33E+00	3.792E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.20	1.30	8.159E+11	4.601E+10	2.95E+00	1.357E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.30	1.40	7.699E+11	4.028E+10	2.75E+00	1.108E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.40	1.60	7.296E+11	6.733E+10	2.51E+00	1.689E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.60	1.80	6.622E+11	5.397E+10	2.24E+00	1.211E+11	2.89E+00	1.559E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
1.80	2.00	6.083E+11	4.439E+10	2.03E+00	9.007E+10	2.86E+00	1.270E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
2.00	2.20	5.639E+11	3.730E+10	1.85E+00	6.898E+10	2.67E+00	9.956E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
2.20	2.40	5.266E+11	3.203E+10	1.70E+00	5.460E+10	2.46E+00	7.877E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
2.40	2.60	4.946E+11	2.790E+10	1.58E+00	4.421E+10	2.26E+00	6.303E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
2.60	2.80	4.667E+11	2.458E+10	1.48E+00	3.636E+10	2.07E+00	5.098E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
2.80	3.00	4.421E+11	2.283E+10	1.38E+00	3.161E+10	1.92E+00	4.381E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
3.00	3.20	4.193E+11	2.028E+10	1.30E+00	2.647E+10	1.78E+00	3.609E+10	1.34E+00	2.721E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000
3.20	3.40	3.990E+11	1.914E+10	1.23E+00	2.363E+10	1.65E+00	3.157E+10	1.47E+00	2.822E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000

3.40	3.60	3.798E+11	1.739E+10	1.17E+00	2.035E+10	1.54E+00	2.678E+10	1.49E+00	2.600E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	3.624E+11	1.590E+10	1.11E+00	1.765E+10	1.44E+00	2.297E+10	1.48E+00	2.345E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	3.465E+11	1.444E+10	1.05E+00	1.523E+10	1.36E+00	1.963E+10	1.43E+00	2.072E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.321E+11	1.316E+10	1.01E+00	1.328E+10	1.28E+00	1.691E+10	1.38E+00	1.816E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	3.189E+11	1.206E+10	9.69E-01	1.169E+10	1.21E+00	1.465E+10	1.32E+00	1.598E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	3.069E+11	1.121E+10	9.32E-01	1.045E+10	1.15E+00	1.289E+10	1.27E+00	1.429E+10	7.38E-01	8.271E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	2.957E+11	1.045E+10	8.97E-01	9.381E+09	1.09E+00	1.145E+10	1.22E+00	1.275E+10	8.97E-01	9.372E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	2.852E+11	1.916E+10	8.51E-01	1.631E+10	1.02E+00	1.963E+10	1.14E+00	2.192E+10	9.66E-01	1.851E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	2.661E+11	1.721E+10	7.97E-01	1.371E+10	9.43E-01	1.623E+10	1.05E+00	1.814E+10	9.84E-01	1.692E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	2.488E+11	1.548E+10	7.50E-01	1.161E+10	8.74E-01	1.354E+10	9.75E-01	1.509E+10	9.57E-01	1.482E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	2.334E+11	1.368E+10	7.10E-01	9.714E+09	8.15E-01	1.116E+10	9.08E-01	1.242E+10	9.16E-01	1.253E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	2.197E+11	1.232E+10	6.76E-01	8.329E+09	7.65E-01	9.424E+09	8.48E-01	1.045E+10	8.71E-01	1.074E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	2.074E+11	1.111E+10	6.45E-01	7.170E+09	7.22E-01	8.019E+09	7.97E-01	8.851E+09	8.27E-01	9.191E+09	5.75E-01	6.388E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	1.963E+11	1.031E+10	6.18E-01	6.377E+09	6.84E-01	7.057E+09	7.52E-01	7.753E+09	7.85E-01	8.098E+09	6.38E-01	6.581E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	1.859E+11	9.280E+09	5.94E-01	5.517E+09	6.52E-01	6.051E+09	7.12E-01	6.612E+09	7.46E-01	6.927E+09	6.56E-01	6.092E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	1.767E+11	2.011E+10	5.60E-01	1.126E+10	6.06E-01	1.218E+10	6.55E-01	1.318E+10	6.88E-01	1.383E+10	6.45E-01	1.297E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	1.565E+11	1.639E+10	5.19E-01	8.498E+09	5.52E-01	9.047E+09	5.89E-01	9.660E+09	6.18E-01	1.013E+10	6.10E-01	9.998E+09	4.17E-01	6.828E+09	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	1.402E+11	1.413E+10	4.88E-01	6.896E+09	5.12E-01	7.235E+09	5.40E-01	7.630E+09	5.64E-01	7.968E+09	5.67E-01	8.018E+09	4.85E-01	6.858E+09	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	1.260E+11	1.206E+10	4.66E-01	5.614E+09	4.83E-01	5.819E+09	5.03E-01	6.060E+09	5.21E-01	6.283E+09	5.28E-01	6.368E+09	4.86E-01	5.862E+09	2.47E-01	2.982E+09	0.00E+00	0.00
12.00	13.00	1.140E+11	1.034E+10	4.46E-01	4.612E+09	4.59E-01	4.747E+09	4.75E-01	4.907E+09	4.88E-01	5.041E+09	4.95E-01	5.113E+09	4.71E-01	4.866E+09	3.70E-01	3.831E+09	0.00E+00	0.00
13.00	14.00	1.036E+11	8.735E+09	4.29E-01	3.748E+09	4.39E-01	3.835E+09	4.52E-01	3.944E+09	4.62E-01	4.032E+09	4.67E-01	4.075E+09	4.52E-01	3.944E+09	3.92E-01	3.424E+09	0.00E+00	0.00
14.00	15.00	9.489E+10	7.776E+09	4.14E-01	3.216E+09	4.22E-01	3.282E+09	4.32E-01	3.359E+09	4.40E-01	3.422E+09	4.44E-01	3.453E+09	4.33E-01	3.367E+09	3.93E-01	3.056E+09	0.00E+00	0.00
15.00	16.00	8.712E+10	6.696E+09	4.00E-01	2.675E+09	4.07E-01	2.725E+09	4.15E-01	2.779E+09	4.22E-01	2.823E+09	4.25E-01	2.846E+09	4.16E-01	2.786E+09	3.87E-01	2.592E+09	0.00E+00	0.00
16.00	18.00	8.042E+10	1.137E+10	3.82E-01	4.337E+09	3.88E-01	4.405E+09	3.94E-01	4.479E+09	3.99E-01	4.536E+09	4.02E-01	4.570E+09	3.96E-01	4.496E+09	3.75E-01	4.258E+09	0.00E+00	0.00
18.00	20.00	6.906E+10	9.057E+09	3.61E-01	3.266E+09	3.65E-01	3.306E+09	3.70E-01	3.352E+09	3.74E-01	3.383E+09	3.77E-01	3.411E+09	3.73E-01	3.374E+09	3.58E-01	3.243E+09	2.60E-01	2.35
20.00	22.00	6.000E+10	7.441E+09	3.47E-01	2.579E+09	3.49E-01	2.597E+09	3.52E-01	2.620E+09	3.54E-01	2.635E+09	3.56E-01	2.646E+09	3.53E-01	2.627E+09	3.43E-01	2.553E+09	2.80E-01	2.08
22.00	24.00	5.256E+10	6.256E+09	3.38E-01	2.115E+09	3.39E-01	2.121E+09	3.41E-01	2.131E+09	3.41E-01	2.134E+09	3.40E-01	2.127E+09	3.37E-01	2.109E+09	3.30E-01	2.062E+09	2.85E-01	1.78
24.00	26.00	4.630E+10	5.193E+09	3.31E-01	1.716E+09	3.32E-01	1.722E+09	3.32E-01	1.724E+09	3.32E-01	1.724E+09	3.30E-01	1.714E+09	3.25E-01	1.690E+09	3.18E-01	1.654E+09	2.85E-01	1.47
26.00	28.00	4.111E+10	4.372E+09	3.24E-01	1.417E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.23E-01	1.410E+09	3.18E-01	1.388E+09	3.11E-01	1.358E+09	2.82E-01	1.23
28.00	30.00	3.674E+10	3.814E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.16E-01	1.205E+09	3.11E-01	1.186E+09	3.05E-01	1.161E+09	2.79E-01	1.06
30.00	34.00	3.292E+10	5.993E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.10E-01	1.858E+09	3.08E-01	1.846E+09	3.04E-01	1.819E+09	2.98E-01	1.783E+09	2.75E-01	1.64
34.00	38.00	2.693E+10	4.589E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.02E-01	1.384E+09	3.00E-01	1.375E+09	2.96E-01	1.356E+09	2.90E-01	1.331E+09	2.72E-01	1.24
38.00	42.00	2.234E+10	3.566E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.95E-01	1.050E+09	2.93E-01	1.043E+09	2.89E-01	1.031E+09	2.85E-01	1.015E+09	2.69E-01	9.59
42.00	46.00	1.878E+10	2.822E+09	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.88E-01	8.129E+08	2.86E-01	8.072E+08	2.84E-01	8.001E+08	2.80E-01	7.903E+08	2.67E-01	7.53
46.00	50.00	1.595E+10	2.260E+09	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.81E-01	6.341E+08	2.79E-01	6.296E+08	2.76E-01	6.228E+08	2.65E-01	5.97
50.00	55.00	1.369E+10	2.255E+09	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.214E+08	2.75E-01	6.191E+08	2.73E-01	6.157E+08	2.71E-01	6.101E+08	2.62E-01	5.89
55.00	60.00	1.144E+10	1.814E+09	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.888E+08	2.69E-01	4.870E+08	2.68E-01	4.852E+08	2.66E-01	4.816E+08	2.58E-01	4.68
60.00	65.00	9.624E+09	1.432E+09	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.64E-01	3.775E+08	2.63E-01	3.760E+08	2.61E-01	3.732E+08	2.55E-01	3.64
65.00	70.00	8.192E+09	1.159E+09	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.008E+08	2.59E-01	3.002E+08	2.58E-01	2.991E+08	2.56E-01	2.968E+08	2.52E-01	2.91
70.00	80.00	7.033E+09	1.736E+09	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.54E-01	4.410E+08	2.54E-01	4.401E+08	2.53E-01	4.384E+08	2.51E-01	4.358E+08	2.47E-01	4.28
80.00	90.00	5.298E+09	1.215E+09	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.48E-01	3.007E+08	2.47E-01	3.001E+08	2.46E-01	2.989E+08	2.43E-01	2.94
90.00	100.00	4.083E+09	8.637E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.44E-01	2.103E+08	2.43E-01	2.099E+08	2.43E-01	2.095E+08	2.40E-01	2.06
100.00	130.00	3.219E+09	1.448E+09	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.458E+08	2.38E-01	3.443E+08	2.38E-01	3.443E+08	2.35E-01	3.40
130.00	160.00	1.771E+09	6.992E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.32E-01	1.623E+08	2.32E-01	1.620E+08	2.32E-01	1.620E+08	2.30E-01	1.60
160.00	200.00	1.071E+09	4.569E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.25E-01	1.02
Equiv. 10 MeV Proton Fluence (cm^{-2})					5.506E+13		9.778E+11		3.656E+11		1.950E+11		9.804E+10		6.035E+10		4.103E+10		1.84
P_{max} Equiv. 1 MeV Electron Fluence (cm^{-2})					6.248E+16		1.110E+15		4.149E+14		2.213E+14		1.113E+14		6.849E+13		4.656E+13		2.09

Solar protons for V_{oc} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	4.895E+12	1.926E+12	1.85E+01	3.555E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.20	0.30	2.969E+12	7.532E+11	1.22E+01	9.156E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.30	0.40	2.216E+12	4.155E+11	9.15E+00	3.801E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.40	0.60	1.801E+12	4.564E+11	6.81E+00	3.108E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.60	0.80	1.344E+12	2.523E+11	5.00E+00	1.262E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.80	1.00	1.092E+12	1.621E+11	3.98E+00	6.459E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.00	1.20	9.299E+11	1.140E+11	3.33E+00	3.792E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.20	1.30	8.159E+11	4.601E+10	2.95E+00	1.357E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.30	1.40	7.699E+11	4.028E+10	2.75E+00	1.108E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.40	1.60	7.296E+11	6.733E+10	2.51E+00	1.689E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.60	1.80	6.622E+11	5.397E+10	2.24E+00	1.211E+11	2.89E+00	1.559E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.80	2.00	6.083E+11	4.439E+10	2.03E+00	9.007E+10	2.86E+00	1.270E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.00	2.20	5.639E+11	3.730E+10	1.85E+00	6.898E+10	2.67E+00	9.956E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.20	2.40	5.266E+11	3.203E+10	1.70E+00	5.460E+10	2.46E+00	7.877E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.40	2.60	4.946E+11	2.790E+10	1.58E+00	4.421E+10	2.26E+00	6.303E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.60	2.80	4.667E+11	2.458E+10	1.48E+00	3.636E+10	2.07E+00	5.098E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00

2.80	3.00	4.421E+11	2.283E+10	1.38E+00	3.161E+10	1.92E+00	4.381E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.00	3.20	4.193E+11	2.028E+10	1.30E+00	2.647E+10	1.78E+00	3.609E+10	1.34E+00	2.721E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.20	3.40	3.990E+11	1.914E+10	1.23E+00	2.363E+10	1.65E+00	3.157E+10	1.47E+00	2.822E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.40	3.60	3.798E+11	1.739E+10	1.17E+00	2.035E+10	1.54E+00	2.678E+10	1.49E+00	2.600E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	3.624E+11	1.590E+10	1.11E+00	1.765E+10	1.44E+00	2.297E+10	1.48E+00	2.345E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	3.465E+11	1.444E+10	1.05E+00	1.523E+10	1.36E+00	1.963E+10	1.43E+00	2.072E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.321E+11	1.316E+10	1.01E+00	1.328E+10	1.28E+00	1.691E+10	1.38E+00	1.816E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	3.189E+11	1.206E+10	9.69E-01	1.169E+10	1.21E+00	1.465E+10	1.32E+00	1.598E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	3.069E+11	1.121E+10	9.32E-01	1.045E+10	1.15E+00	1.289E+10	1.27E+00	1.429E+10	7.38E-01	8.271E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	2.957E+11	1.045E+10	8.97E-01	9.381E+09	1.09E+00	1.145E+10	1.22E+00	1.275E+10	8.97E-01	9.372E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	2.852E+11	1.916E+10	8.51E-01	1.631E+10	1.02E+00	1.963E+10	1.14E+00	2.192E+10	9.66E-01	1.851E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	2.661E+11	1.721E+10	7.97E-01	1.371E+10	9.43E-01	1.623E+10	1.05E+00	1.814E+10	9.84E-01	1.692E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	2.488E+11	1.548E+10	7.50E-01	1.161E+10	8.74E-01	1.354E+10	9.75E-01	1.509E+10	9.57E-01	1.482E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	2.334E+11	1.368E+10	7.10E-01	9.714E+09	8.15E-01	1.116E+10	9.08E-01	1.242E+10	9.16E-01	1.253E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	2.197E+11	1.232E+10	6.76E-01	8.329E+09	7.65E-01	9.424E+09	8.48E-01	1.045E+10	8.71E-01	1.074E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	2.074E+11	1.111E+10	6.45E-01	7.170E+09	7.22E-01	8.019E+09	7.97E-01	8.851E+09	8.27E-01	9.191E+09	5.75E-01	6.388E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	1.963E+11	1.031E+10	6.18E-01	6.377E+09	6.84E-01	7.057E+09	7.52E-01	7.753E+09	7.85E-01	8.098E+09	6.38E-01	6.581E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	1.859E+11	9.280E+09	5.94E-01	5.517E+09	6.52E-01	6.051E+09	7.12E-01	6.612E+09	7.46E-01	6.927E+09	6.56E-01	6.092E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	1.767E+11	2.011E+10	5.60E-01	1.126E+10	6.06E-01	1.218E+10	6.55E-01	1.318E+10	6.88E-01	1.383E+10	6.45E-01	1.297E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	1.565E+11	1.639E+10	5.19E-01	8.498E+09	5.52E-01	9.047E+09	5.89E-01	9.660E+09	6.18E-01	1.013E+10	6.10E-01	9.998E+09	4.17E-01	6.828E+09	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	1.402E+11	1.413E+10	4.88E-01	6.896E+09	5.12E-01	7.235E+09	5.40E-01	7.630E+09	5.64E-01	7.968E+09	5.67E-01	8.018E+09	4.85E-01	6.858E+09	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	1.260E+11	1.206E+10	4.66E-01	5.614E+09	4.83E-01	5.819E+09	5.03E-01	6.060E+09	5.21E-01	6.283E+09	5.28E-01	6.368E+09	4.86E-01	5.862E+09	2.47E-01	2.982E+09	0.00E+00	0.00
12.00	13.00	1.140E+11	1.034E+10	4.46E-01	4.612E+09	4.59E-01	4.747E+09	4.75E-01	4.907E+09	4.88E-01	5.041E+09	4.95E-01	5.113E+09	4.71E-01	4.866E+09	3.70E-01	3.831E+09	0.00E+00	0.00
13.00	14.00	1.036E+11	8.735E+09	4.29E-01	3.748E+09	4.39E-01	3.835E+09	4.52E-01	3.944E+09	4.62E-01	4.032E+09	4.67E-01	4.075E+09	4.52E-01	3.944E+09	3.92E-01	3.424E+09	0.00E+00	0.00
14.00	15.00	9.489E+10	7.776E+09	4.14E-01	3.216E+09	4.22E-01	3.282E+09	4.32E-01	3.359E+09	4.40E-01	3.422E+09	4.44E-01	3.453E+09	4.33E-01	3.367E+09	3.93E-01	3.056E+09	0.00E+00	0.00
15.00	16.00	8.712E+10	6.696E+09	4.00E-01	2.675E+09	4.07E-01	2.725E+09	4.15E-01	2.779E+09	4.22E-01	2.823E+09	4.25E-01	2.846E+09	4.16E-01	2.786E+09	3.87E-01	2.592E+09	0.00E+00	0.00
16.00	18.00	8.042E+10	1.137E+10	3.82E-01	4.337E+09	3.88E-01	4.405E+09	3.94E-01	4.479E+09	3.99E-01	4.536E+09	4.02E-01	4.570E+09	3.96E-01	4.496E+09	3.75E-01	4.258E+09	0.00E+00	0.00
18.00	20.00	6.906E+10	9.057E+09	3.61E-01	3.266E+09	3.65E-01	3.306E+09	3.70E-01	3.352E+09	3.74E-01	3.383E+09	3.77E-01	3.411E+09	3.73E-01	3.374E+09	3.58E-01	3.243E+09	2.60E-01	2.35
20.00	22.00	6.000E+10	7.441E+09	3.47E-01	2.579E+09	3.49E-01	2.597E+09	3.52E-01	2.620E+09	3.54E-01	2.635E+09	3.56E-01	2.646E+09	3.53E-01	2.627E+09	3.43E-01	2.553E+09	2.80E-01	2.08
22.00	24.00	5.256E+10	6.256E+09	3.38E-01	2.115E+09	3.39E-01	2.121E+09	3.41E-01	2.131E+09	3.41E-01	2.134E+09	3.40E-01	2.127E+09	3.37E-01	2.109E+09	3.30E-01	2.062E+09	2.85E-01	1.78
24.00	26.00	4.630E+10	5.193E+09	3.31E-01	1.716E+09	3.32E-01	1.722E+09	3.32E-01	1.724E+09	3.32E-01	1.724E+09	3.30E-01	1.714E+09	3.25E-01	1.690E+09	3.18E-01	1.654E+09	2.85E-01	1.47
26.00	28.00	4.111E+10	4.372E+09	3.24E-01	1.417E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.23E-01	1.410E+09	3.18E-01	1.388E+09	3.11E-01	1.358E+09	2.82E-01	1.23
28.00	30.00	3.674E+10	3.814E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.16E-01	1.205E+09	3.11E-01	1.186E+09	3.05E-01	1.161E+09	2.79E-01	1.06
30.00	34.00	3.292E+10	5.993E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.10E-01	1.858E+09	3.08E-01	1.846E+09	3.04E-01	1.819E+09	2.98E-01	1.783E+09	2.75E-01	1.64
34.00	38.00	2.693E+10	4.589E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.02E-01	1.384E+09	3.00E-01	1.375E+09	2.96E-01	1.356E+09	2.90E-01	1.331E+09	2.72E-01	1.24
38.00	42.00	2.234E+10	3.566E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.95E-01	1.050E+09	2.93E-01	1.043E+09	2.89E-01	1.031E+09	2.85E-01	1.015E+09	2.69E-01	9.59
42.00	46.00	1.878E+10	2.822E+09	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.88E-01	8.129E+08	2.86E-01	8.072E+08	2.84E-01	8.001E+08	2.80E-01	7.903E+08	2.67E-01	7.53
46.00	50.00	1.595E+10	2.260E+09	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.81E-01	6.341E+08	2.79E-01	6.296E+08	2.76E-01	6.228E+08	2.65E-01	5.97
50.00	55.00	1.369E+10	2.255E+09	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.214E+08	2.75E-01	6.191E+08	2.73E-01	6.157E+08	2.71E-01	6.101E+08	2.62E-01	5.89
55.00	60.00	1.144E+10	1.814E+09	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.888E+08	2.69E-01	4.870E+08	2.68E-01	4.852E+08	2.66E-01	4.816E+08	2.58E-01	4.68
60.00	65.00	9.624E+09	1.432E+09	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.64E-01	3.775E+08	2.63E-01	3.760E+08	2.61E-01	3.732E+08	2.55E-01	3.64
65.00	70.00	8.192E+09	1.159E+09	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.008E+08	2.59E-01	3.002E+08	2.58E-01	2.991E+08	2.56E-01	2.968E+08	2.52E-01	2.91
70.00	80.00	7.033E+09	1.736E+09	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.54E-01	4.410E+08	2.54E-01	4.401E+08	2.53E-01	4.384E+08	2.51E-01	4.358E+08	2.47E-01	4.28
80.00	90.00	5.298E+09	1.215E+09	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.48E-01	3.007E+08	2.47E-01	3.001E+08	2.46E-01	2.989E+08	2.43E-01	2.94
90.00	100.00	4.083E+09	8.637E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.44E-01	2.103E+08	2.43E-01	2.099E+08	2.43E-01	2.095E+08	2.40E-01	2.06
100.00	130.00	3.219E+09	1.448E+09	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.458E+08	2.38E-01	3.443E+08	2.38E-01	3.443E+08	2.35E-01	3.40
130.00	160.00	1.771E+09	6.992E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.32E-01	1.623E+08	2.32E-01	1.620E+08	2.32E-01	1.620E+08	2.30E-01	1.60
160.00	200.00	1.071E+09	4.569E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.25E-01	1.02
Equiv. 10 MeV Proton Fluence (cm^{-2})					5.506E+13		9.778E+11		3.656E+11		1.950E+11		9.804E+10		6.035E+10		4.103E+10		1.84
V_{OC} Equiv. 1 MeV Electron Fluence (cm^{-2})					1.577E+17		2.801E+15		1.047E+15		5.587E+14		2.808E+14		1.729E+14		1.175E+14		5.28

Solar protons for I_{sc} (integration restricted to the input energy range). D(E,..) values are coarse averages introduced into the table to simp the output information; the accuracy of the results should not be affected by this simplification.

				0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micr	
Energy Int. (MeV)		Int. Fluence (cm^{-2})	Diff. Fluence (cm^{-2})	D(E, 0.0)	Eq. Fluence (cm^{-2})	D(E, 1.0)	Eq. Fluence (cm^{-2})	D(E, 3.0)	Eq. Fluence (cm^{-2})	D(E, 6.0)	Eq. Fluence (cm^{-2})	D(E,12.0)	Eq. Fluence (cm^{-2})	D(E,20.0)	Eq. Fluence (cm^{-2})	D(E,30.0)	Eq. Fluence (cm^{-2})	D(E,60.0)	Eq. Flu (cm
0.10	0.20	4.895E+12	1.926E+12	1.85E+01	3.555E+13	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.20	0.30	2.969E+12	7.532E+11	1.22E+01	9.156E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.30	0.40	2.216E+12	4.155E+11	9.15E+00	3.801E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.40	0.60	1.801E+12	4.564E+11	6.81E+00	3.108E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.60	0.80	1.344E+12	2.523E+11	5.00E+00	1.262E+12	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
0.80	1.00	1.092E+12	1.621E+11	3.98E+00	6.459E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.00	1.20	9.299E+11	1.140E+11	3.33E+00	3.792E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.20	1.30	8.159E+11	4.601E+10	2.95E+00	1.357E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.30	1.40	7.699E+11	4.028E+10	2.75E+00	1.108E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.40	1.60	7.296E+11	6.733E+10	2.51E+00	1.689E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.60	1.80	6.622E+11	5.397E+10	2.24E+00	1.211E+11	2.89E+00	1.559E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
1.80	2.00	6.083E+11	4.439E+10	2.03E+00	9.007E+10	2.86E+00	1.270E+11	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.00	2.20	5.639E+11	3.730E+10	1.85E+00	6.898E+10	2.67E+00	9.956E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.20	2.40	5.266E+11	3.203E+10	1.70E+00	5.460E+10	2.46E+00	7.877E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00

2.40	2.60	4.946E+11	2.790E+10	1.58E+00	4.421E+10	2.26E+00	6.303E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.60	2.80	4.667E+11	2.458E+10	1.48E+00	3.636E+10	2.07E+00	5.098E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
2.80	3.00	4.421E+11	2.283E+10	1.38E+00	3.161E+10	1.92E+00	4.381E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.00	3.20	4.193E+11	2.028E+10	1.30E+00	2.647E+10	1.78E+00	3.609E+10	1.34E+00	2.721E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.20	3.40	3.990E+11	1.914E+10	1.23E+00	2.363E+10	1.65E+00	3.157E+10	1.47E+00	2.822E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.40	3.60	3.798E+11	1.739E+10	1.17E+00	2.035E+10	1.54E+00	2.678E+10	1.49E+00	2.600E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.60	3.80	3.624E+11	1.590E+10	1.11E+00	1.765E+10	1.44E+00	2.297E+10	1.48E+00	2.345E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
3.80	4.00	3.465E+11	1.444E+10	1.05E+00	1.523E+10	1.36E+00	1.963E+10	1.43E+00	2.072E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.00	4.20	3.321E+11	1.316E+10	1.01E+00	1.328E+10	1.28E+00	1.691E+10	1.38E+00	1.816E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.20	4.40	3.189E+11	1.206E+10	9.69E-01	1.169E+10	1.21E+00	1.465E+10	1.32E+00	1.598E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.40	4.60	3.069E+11	1.121E+10	9.32E-01	1.045E+10	1.15E+00	1.289E+10	1.27E+00	1.429E+10	7.38E-01	8.271E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.60	4.80	2.957E+11	1.045E+10	8.97E-01	9.381E+09	1.09E+00	1.145E+10	1.22E+00	1.275E+10	8.97E-01	9.372E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
4.80	5.20	2.852E+11	1.916E+10	8.51E-01	1.631E+10	1.02E+00	1.963E+10	1.14E+00	2.192E+10	9.66E-01	1.851E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.20	5.60	2.661E+11	1.721E+10	7.97E-01	1.371E+10	9.43E-01	1.623E+10	1.05E+00	1.814E+10	9.84E-01	1.692E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
5.60	6.00	2.488E+11	1.548E+10	7.50E-01	1.161E+10	8.74E-01	1.354E+10	9.75E-01	1.509E+10	9.57E-01	1.482E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.00	6.40	2.334E+11	1.368E+10	7.10E-01	9.714E+09	8.15E-01	1.116E+10	9.08E-01	1.242E+10	9.16E-01	1.253E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.40	6.80	2.197E+11	1.232E+10	6.76E-01	8.329E+09	7.65E-01	9.424E+09	8.48E-01	1.045E+10	8.71E-01	1.074E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
6.80	7.20	2.074E+11	1.111E+10	6.45E-01	7.170E+09	7.22E-01	8.019E+09	7.97E-01	8.851E+09	8.27E-01	9.191E+09	5.75E-01	6.388E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.20	7.60	1.963E+11	1.031E+10	6.18E-01	6.377E+09	6.84E-01	7.057E+09	7.52E-01	7.753E+09	7.85E-01	8.098E+09	6.38E-01	6.581E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
7.60	8.00	1.859E+11	9.280E+09	5.94E-01	5.517E+09	6.52E-01	6.051E+09	7.12E-01	6.612E+09	7.46E-01	6.927E+09	6.56E-01	6.092E+09	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
8.00	9.00	1.767E+11	2.011E+10	5.60E-01	1.126E+10	6.06E-01	1.218E+10	6.55E-01	1.318E+10	6.88E-01	1.383E+10	6.45E-01	1.297E+10	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00
9.00	10.00	1.565E+11	1.639E+10	5.19E-01	8.498E+09	5.52E-01	9.047E+09	5.89E-01	9.660E+09	6.18E-01	1.013E+10	6.10E-01	9.998E+09	4.17E-01	6.828E+09	0.00E+00	0.000E+00	0.00E+00	0.00
10.00	11.00	1.402E+11	1.413E+10	4.88E-01	6.896E+09	5.12E-01	7.235E+09	5.40E-01	7.630E+09	5.64E-01	7.968E+09	5.67E-01	8.018E+09	4.85E-01	6.858E+09	0.00E+00	0.000E+00	0.00E+00	0.00
11.00	12.00	1.260E+11	1.206E+10	4.66E-01	5.614E+09	4.83E-01	5.819E+09	5.03E-01	6.060E+09	5.21E-01	6.283E+09	5.28E-01	6.368E+09	4.86E-01	5.862E+09	2.47E-01	2.982E+09	0.00E+00	0.00
12.00	13.00	1.140E+11	1.034E+10	4.46E-01	4.612E+09	4.59E-01	4.747E+09	4.75E-01	4.907E+09	4.88E-01	5.041E+09	4.95E-01	5.113E+09	4.71E-01	4.866E+09	3.70E-01	3.831E+09	0.00E+00	0.00
13.00	14.00	1.036E+11	8.735E+09	4.29E-01	3.748E+09	4.39E-01	3.835E+09	4.52E-01	3.944E+09	4.62E-01	4.032E+09	4.67E-01	4.075E+09	4.52E-01	3.944E+09	3.92E-01	3.424E+09	0.00E+00	0.00
14.00	15.00	9.489E+10	7.776E+09	4.14E-01	3.216E+09	4.22E-01	3.282E+09	4.32E-01	3.359E+09	4.40E-01	3.422E+09	4.44E-01	3.453E+09	4.33E-01	3.367E+09	3.93E-01	3.056E+09	0.00E+00	0.00
15.00	16.00	8.712E+10	6.696E+09	4.00E-01	2.675E+09	4.07E-01	2.725E+09	4.15E-01	2.779E+09	4.22E-01	2.823E+09	4.25E-01	2.846E+09	4.16E-01	2.786E+09	3.87E-01	2.592E+09	0.00E+00	0.00
16.00	18.00	8.042E+10	1.137E+10	3.82E-01	4.337E+09	3.88E-01	4.405E+09	3.94E-01	4.479E+09	3.99E-01	4.536E+09	4.02E-01	4.570E+09	3.96E-01	4.496E+09	3.75E-01	4.258E+09	0.00E+00	0.00
18.00	20.00	6.906E+10	9.057E+09	3.61E-01	3.266E+09	3.65E-01	3.306E+09	3.70E-01	3.352E+09	3.74E-01	3.383E+09	3.77E-01	3.411E+09	3.73E-01	3.374E+09	3.58E-01	3.243E+09	2.60E-01	2.35
20.00	22.00	6.000E+10	7.441E+09	3.47E-01	2.579E+09	3.49E-01	2.597E+09	3.52E-01	2.620E+09	3.54E-01	2.635E+09	3.56E-01	2.646E+09	3.53E-01	2.627E+09	3.43E-01	2.553E+09	2.80E-01	2.08
22.00	24.00	5.256E+10	6.256E+09	3.38E-01	2.115E+09	3.39E-01	2.121E+09	3.41E-01	2.131E+09	3.41E-01	2.134E+09	3.40E-01	2.127E+09	3.37E-01	2.109E+09	3.30E-01	2.062E+09	2.85E-01	1.78
24.00	26.00	4.630E+10	5.193E+09	3.31E-01	1.716E+09	3.32E-01	1.722E+09	3.32E-01	1.724E+09	3.32E-01	1.724E+09	3.30E-01	1.714E+09	3.25E-01	1.690E+09	3.18E-01	1.654E+09	2.85E-01	1.47
26.00	28.00	4.111E+10	4.372E+09	3.24E-01	1.417E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.25E-01	1.419E+09	3.23E-01	1.410E+09	3.18E-01	1.388E+09	3.11E-01	1.358E+09	2.82E-01	1.23
28.00	30.00	3.674E+10	3.814E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.18E-01	1.213E+09	3.16E-01	1.205E+09	3.11E-01	1.186E+09	3.05E-01	1.161E+09	2.79E-01	1.06
30.00	34.00	3.292E+10	5.993E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.11E-01	1.861E+09	3.10E-01	1.858E+09	3.08E-01	1.846E+09	3.04E-01	1.819E+09	2.98E-01	1.783E+09	2.75E-01	1.64
34.00	38.00	2.693E+10	4.589E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.03E-01	1.388E+09	3.02E-01	1.384E+09	3.00E-01	1.375E+09	2.96E-01	1.356E+09	2.90E-01	1.331E+09	2.72E-01	1.24
38.00	42.00	2.234E+10	3.566E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.96E-01	1.054E+09	2.95E-01	1.050E+09	2.93E-01	1.043E+09	2.89E-01	1.031E+09	2.85E-01	1.015E+09	2.69E-01	9.59
42.00	46.00	1.878E+10	2.822E+09	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.89E-01	8.143E+08	2.88E-01	8.129E+08	2.86E-01	8.072E+08	2.84E-01	8.001E+08	2.80E-01	7.903E+08	2.67E-01	7.53
46.00	50.00	1.595E+10	2.260E+09	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.82E-01	6.375E+08	2.81E-01	6.341E+08	2.79E-01	6.296E+08	2.76E-01	6.228E+08	2.65E-01	5.97
50.00	55.00	1.369E+10	2.255E+09	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.225E+08	2.76E-01	6.214E+08	2.75E-01	6.191E+08	2.73E-01	6.157E+08	2.71E-01	6.101E+08	2.62E-01	5.89
55.00	60.00	1.144E+10	1.814E+09	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.897E+08	2.70E-01	4.888E+08	2.69E-01	4.870E+08	2.68E-01	4.852E+08	2.66E-01	4.816E+08	2.58E-01	4.68
60.00	65.00	9.624E+09	1.432E+09	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.65E-01	3.789E+08	2.64E-01	3.775E+08	2.63E-01	3.760E+08	2.61E-01	3.732E+08	2.55E-01	3.64
65.00	70.00	8.192E+09	1.159E+09	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.014E+08	2.60E-01	3.008E+08	2.59E-01	3.002E+08	2.58E-01	2.991E+08	2.56E-01	2.968E+08	2.52E-01	2.91
70.00	80.00	7.033E+09	1.736E+09	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.55E-01	4.419E+08	2.54E-01	4.410E+08	2.54E-01	4.401E+08	2.53E-01	4.384E+08	2.51E-01	4.358E+08	2.47E-01	4.28
80.00	90.00	5.298E+09	1.215E+09	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.49E-01	3.019E+08	2.48E-01	3.007E+08	2.47E-01	3.001E+08	2.46E-01	2.989E+08	2.43E-01	2.94
90.00	100.00	4.083E+09	8.637E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.45E-01	2.112E+08	2.44E-01	2.103E+08	2.43E-01	2.099E+08	2.43E-01	2.095E+08	2.40E-01	2.06
100.00	130.00	3.219E+09	1.448E+09	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.465E+08	2.39E-01	3.458E+08	2.38E-01	3.443E+08	2.38E-01	3.443E+08	2.35E-01	3.40
130.00	160.00	1.771E+09	6.992E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.33E-01	1.627E+08	2.32E-01	1.623E+08	2.32E-01	1.620E+08	2.32E-01	1.620E+08	2.30E-01	1.60
160.00	200.00	1.071E+09	4.569E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.27E-01	1.038E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.26E-01	1.034E+08	2.25E-01	1.02
Equiv. 10 MeV Proton Fluence (cm^{-2})					5.506E+13		9.778E+11		3.656E+11		1.950E+11		9.804E+10		6.035E+10		4.103E+10		1.84
I_{SC} Equiv. 1 MeV Electron Fluence (cm^{-2})					3.378E+16		5.998E+14		2.243E+14		1.196E+14		6.014E+13		3.702E+13		2.517E+13		1.13